

Turbulence as a fundamental heliophysics process

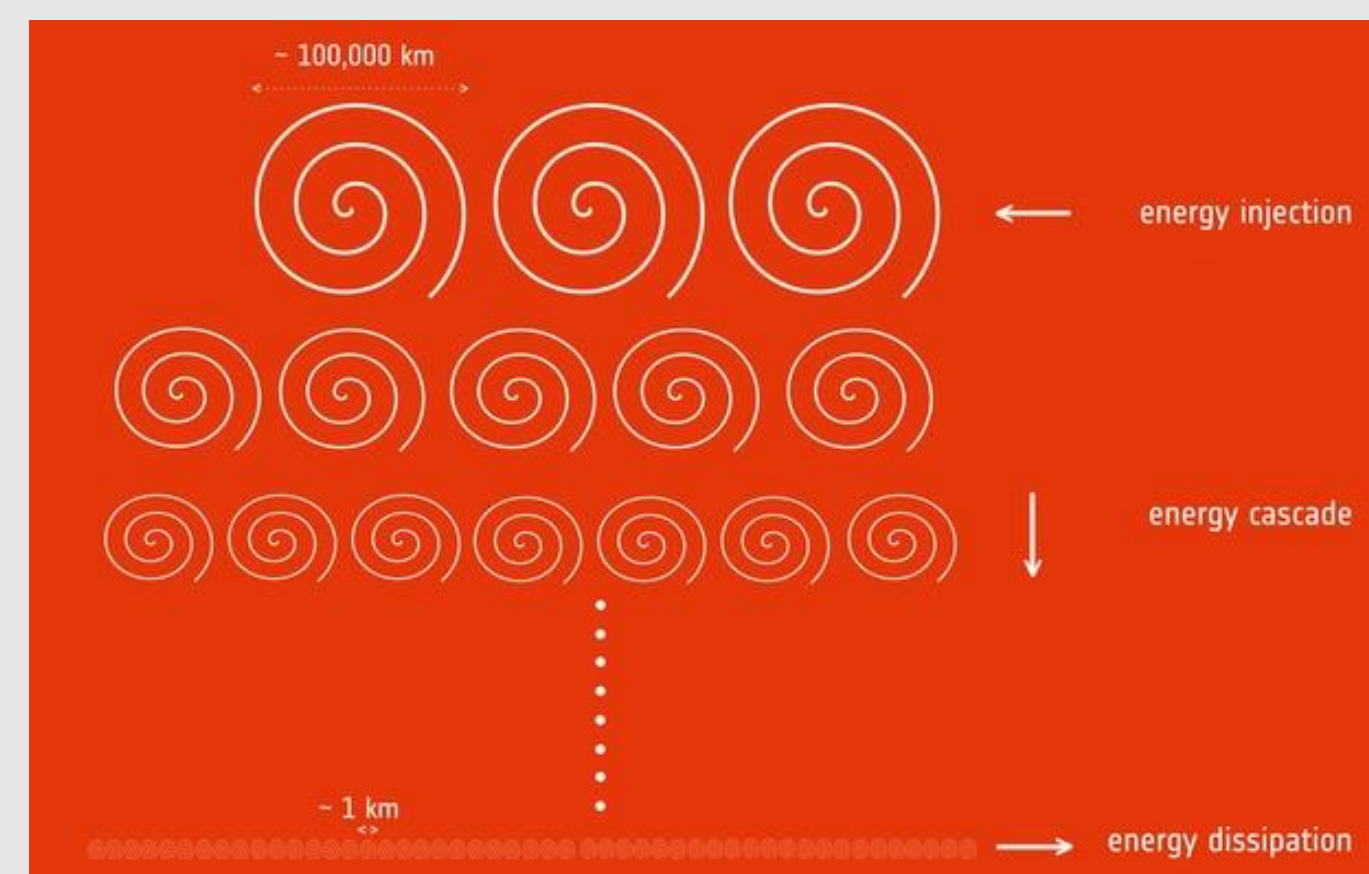
- Turbulence is ubiquitous in space plasmas: in the Sun, solar atmosphere, solar wind, heliosheath, magnetospheres, CMEs, etc.
- Some of its properties may be universal and others not, still to be fully understood
- Can significantly influence other processes (shocks, reconnection, heating, particle acceleration & transport) as well as larger scale system dynamics and structure → broadly important to understand
- Therefore, acts as a connection between different fields of heliophysics

Current state of knowledge

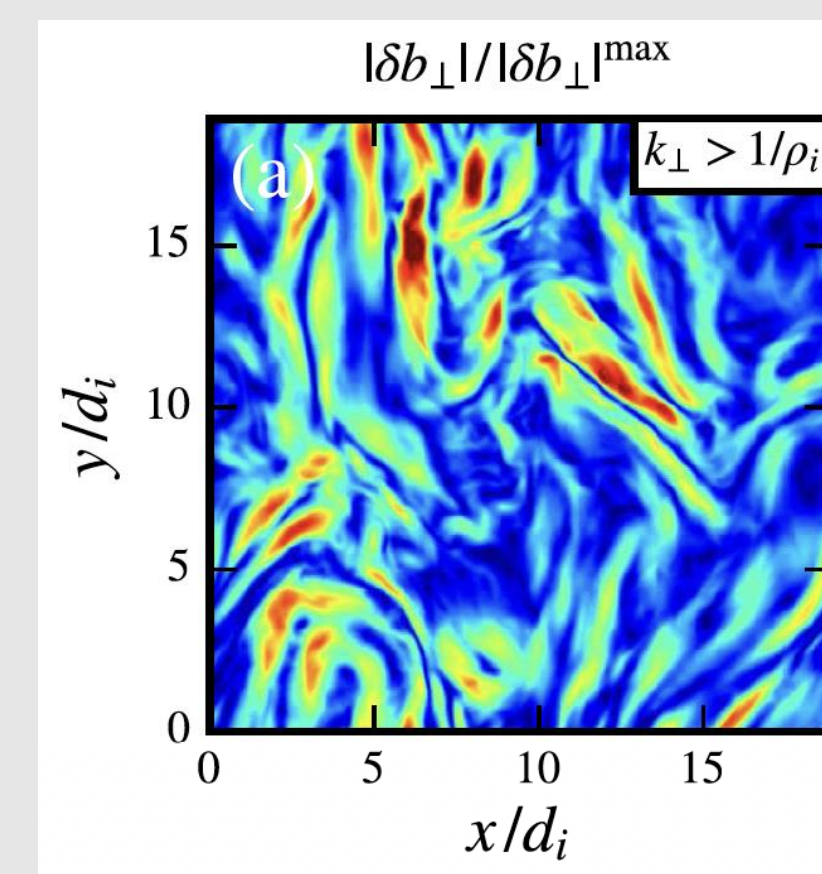
- Basic cascade models describe many features of turbulence, e.g., spectra of fluctuations, field polarisations, plasma structures, intermittency properties, cascade/heating rates
- Incorporation of turbulence into global models, e.g., solar wind models, space weather models, particle transport models, are at different stages but results are generally promising
- However, much remains to be understood: a full picture that captures all features, different parameter regime dependence, universal vs non-universal aspects, driving and dissipation/heating, and all the ways in which it contributes to various aspects of the global structure and dynamics

What is plasma turbulence?

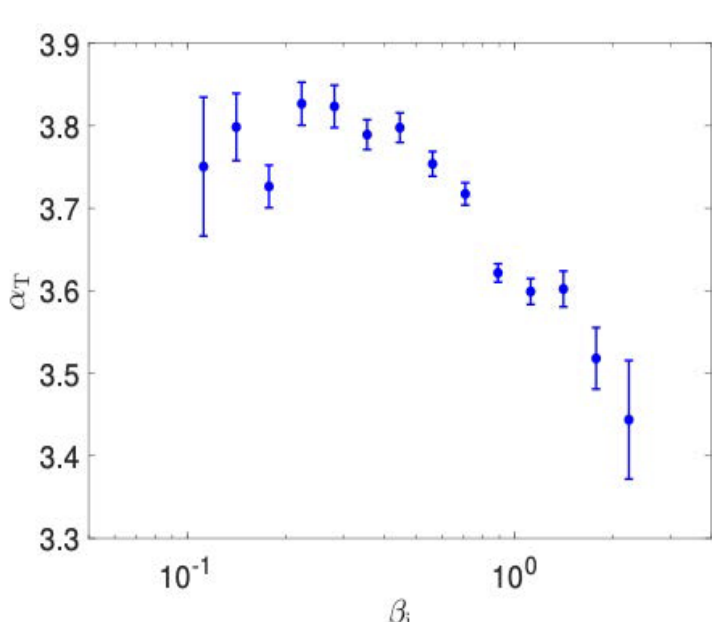
- Broadband multi-scale plasma fluctuations
- Non-linear energy transfer, modelled as energy cascade (large to small scales)
 - Involves a mixture of various linear and non-linear plasma physics
 - High degrees of freedom → statistical approach



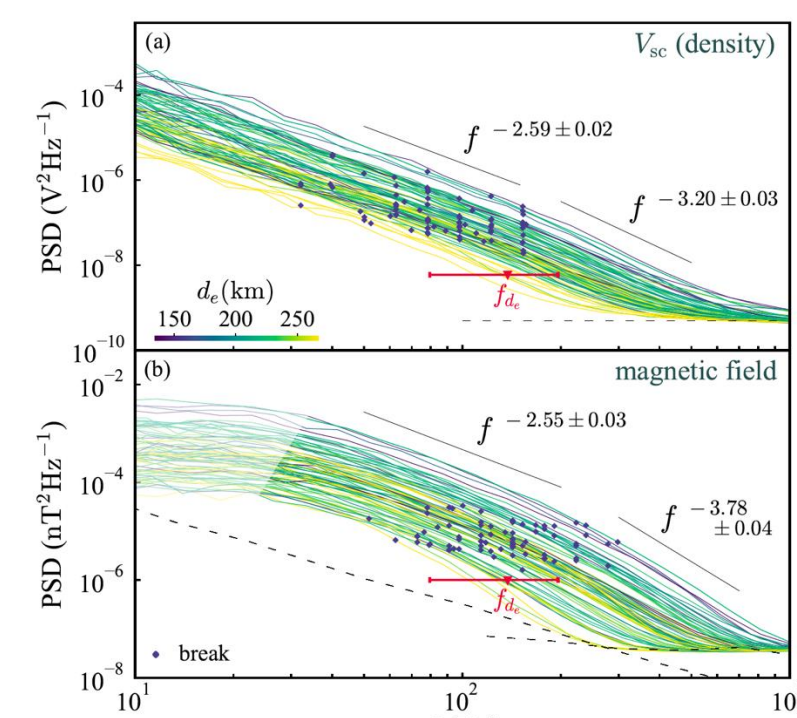
Schematic of energy cascade from large to small scales (Credit: ESA)



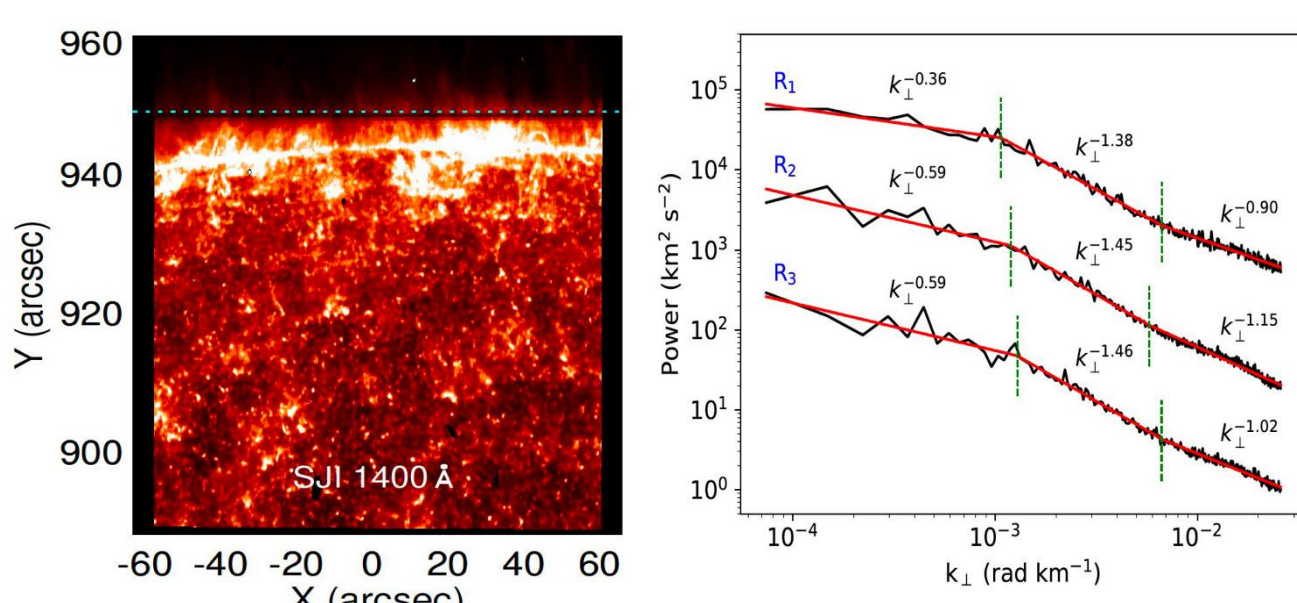
Simulation of turbulent plasma (Grošelj et al. 2019)



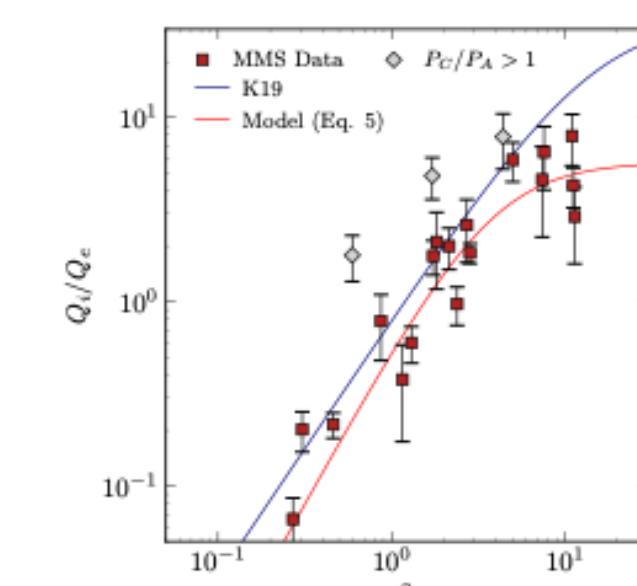
PSP data: transition range spectral index α_T is controlled by ion beta β_i (and σ_e) consistent with helicity barrier (McIntyre et al. 2025)



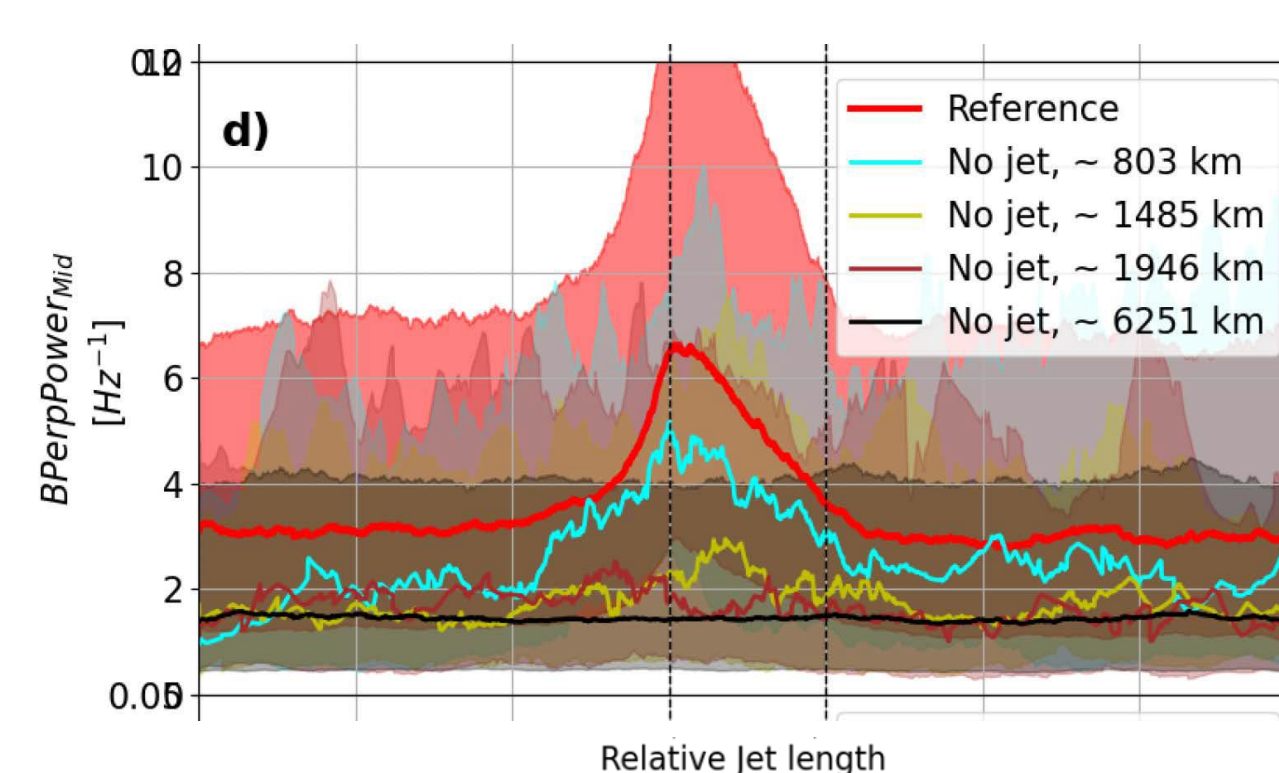
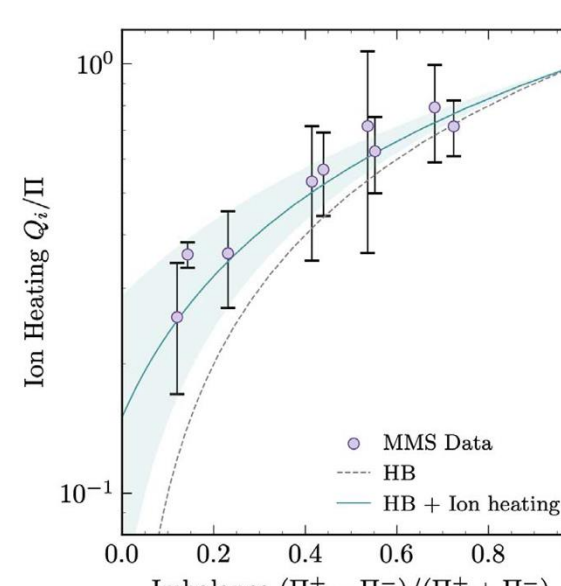
PSP data: sub-electron scales showing transition to electrostatic turbulence consistent with entropy cascade (Mondal et al. 2026)



IRIS data: Alfvénic fluctuations in polar spicules show turbulent spectra and intermittency (Tajfirouze et al. 2026)



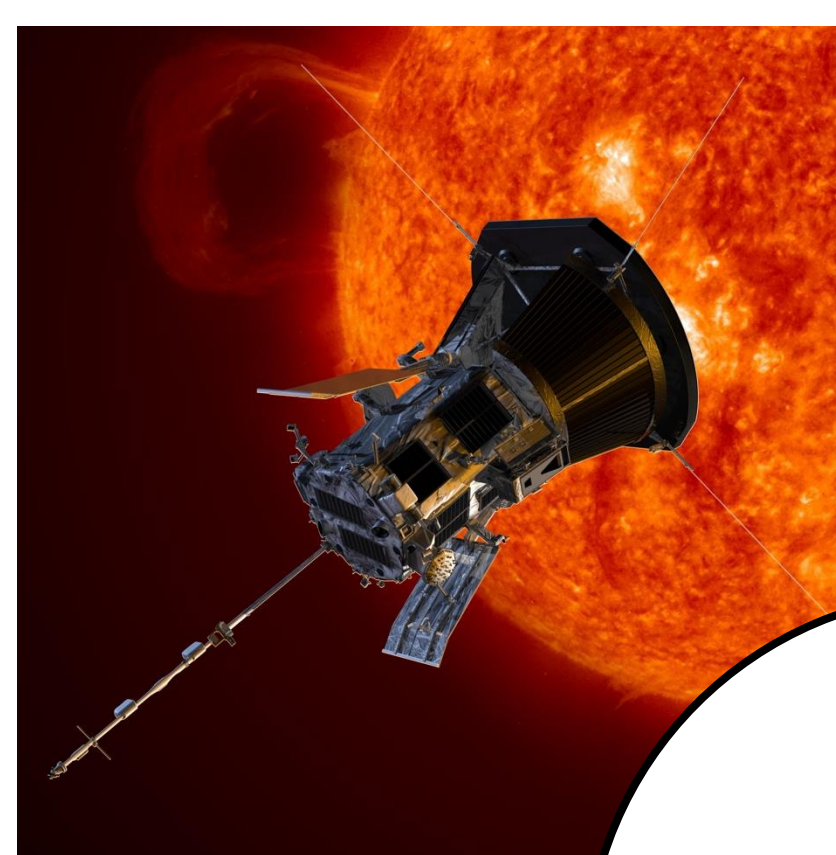
MMS data: ion-electron heating ratio increases with ion beta but saturates from anisotropic viscous heating; it also increases with imbalance consistent with helicity barrier (Manzini et al. 2026a, 2026b)



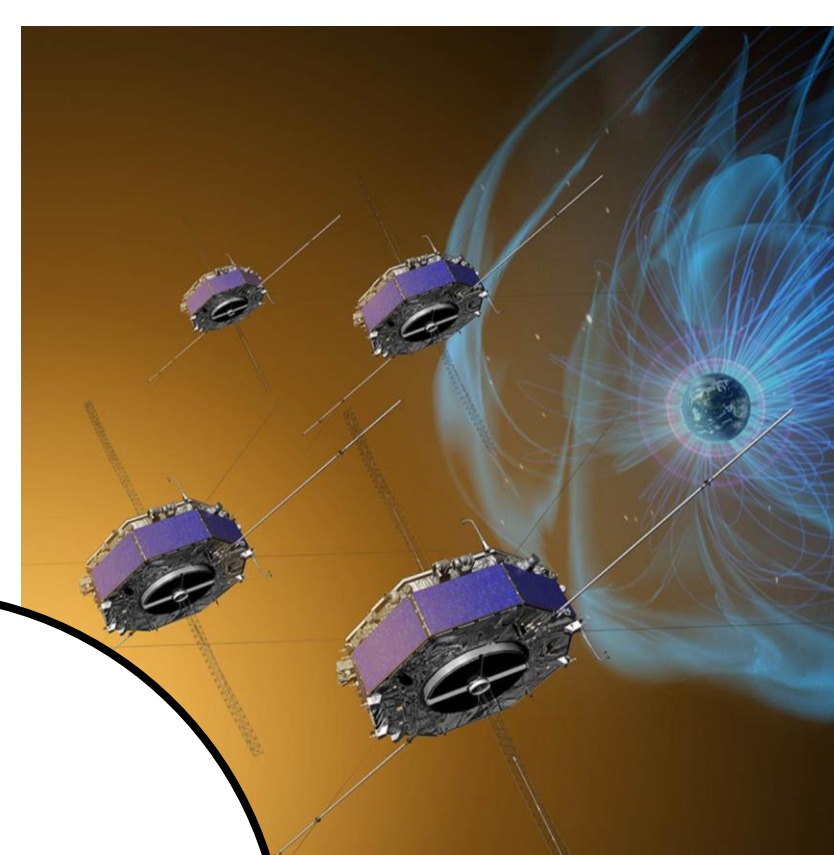
THEMIS data: fluctuation power at different distances away from magnetosheath jets showing enhancements at jet front out to ~1000 km (Koller et al. 2026)

- Investigated turbulence near the Sun, differences to 1 AU, role in coronal heating and solar wind generation
- Found evidence for new “helicity barrier” that modifies cascade and ion heating mechanisms
- Found electrostatic turbulence, consistent with entropy cascade, which can explain electron heating
- Found turbulence in polar spicules, helping to understand its role in solar wind generation

Near-Sun



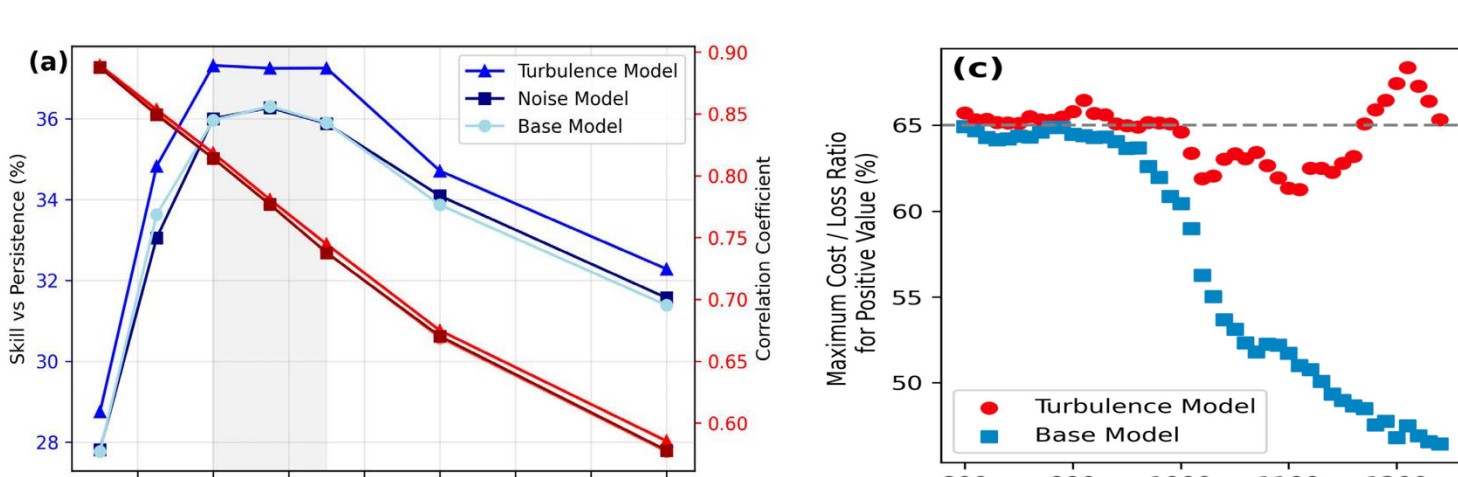
Magnetosheath



- Studied effects of turbulence in Earth's magnetosheath, as a fundamental process and in shaping the environment
- Ion-to-electron heating ratio reveals fundamental mechanisms: Landau damping, anisotropic viscous heating, and helicity barrier
- Enhanced fluctuations found at magnetosheath jet fronts extending ~1000km perpendicular to flow, a possible route for dissipation of jet energy

Plasma Turbulence

- Determined the role of solar wind turbulence in space weather effects and forecasting
- Including solar wind turbulence in AE prediction improves forecast, and maintains economic value for large events
- Time-dependent inner boundary introduces meso-scale solar wind fluctuations and alters stream structure, which significantly impacts CME propagation

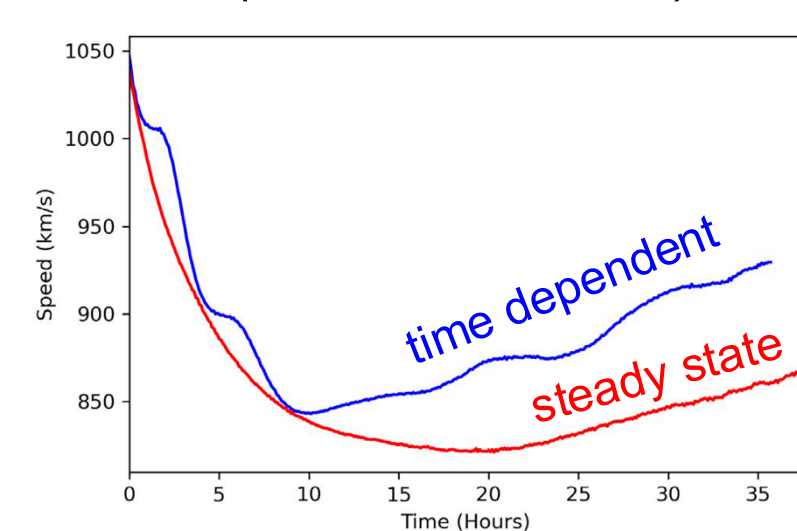


AE index forecast model including solar wind turbulence performs better and maintains economic value for larger events (Waters et al. 2026)

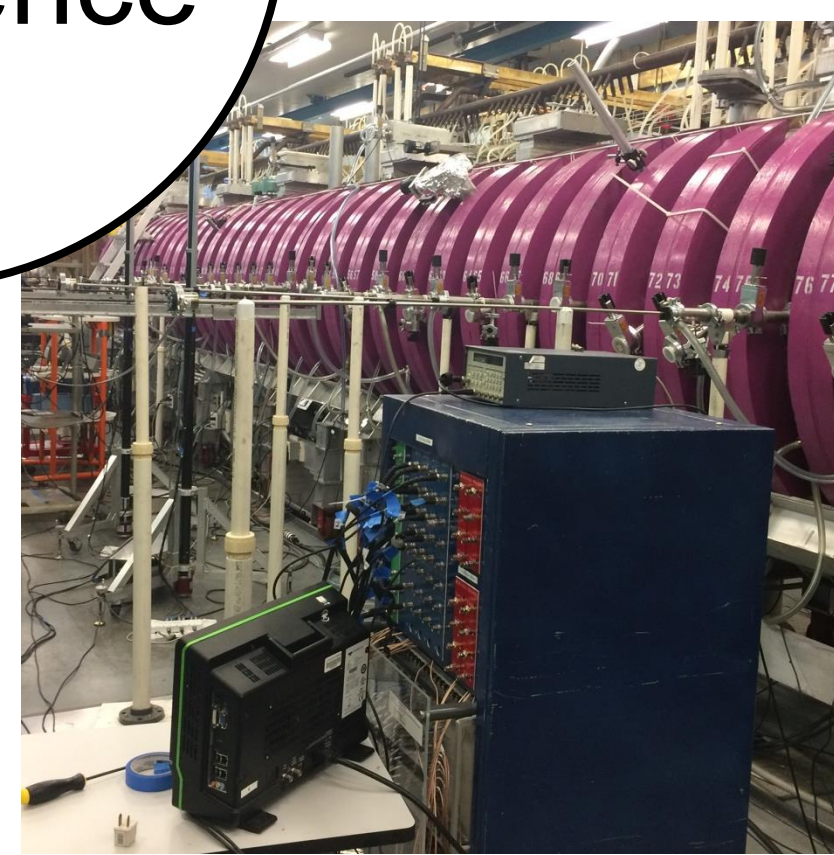


Space Weather

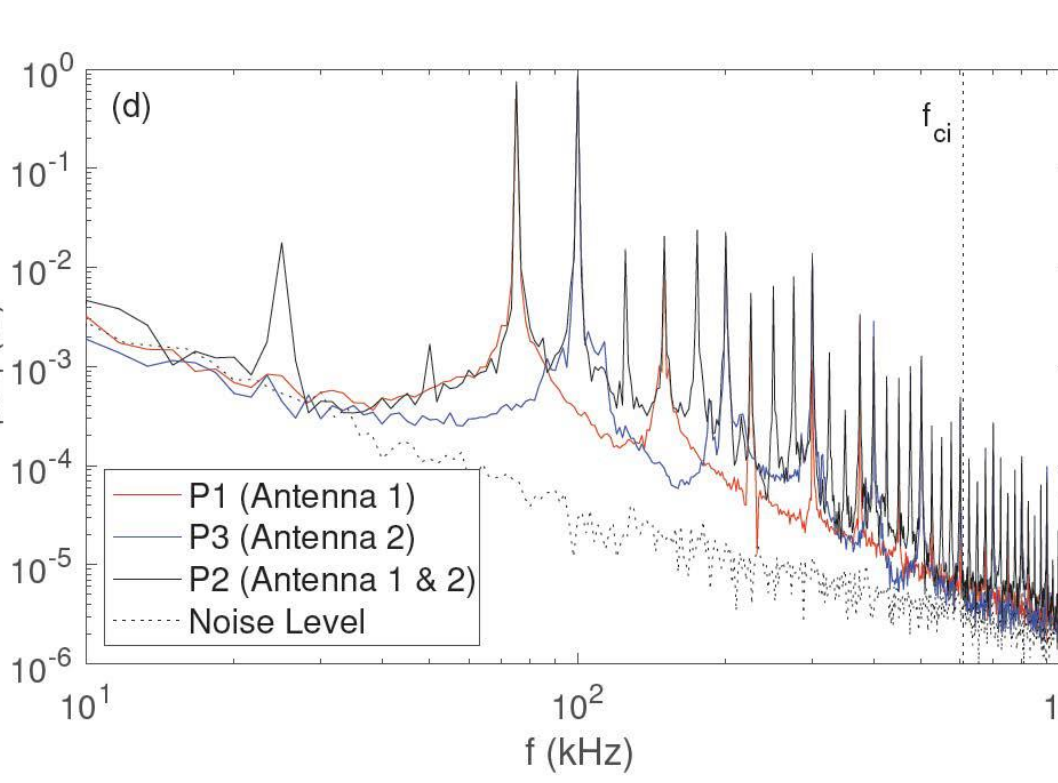
Solar wind model with time-dependent inner boundary produces significant fluctuations that affect CME propagation (Sishtla et al. 2026)



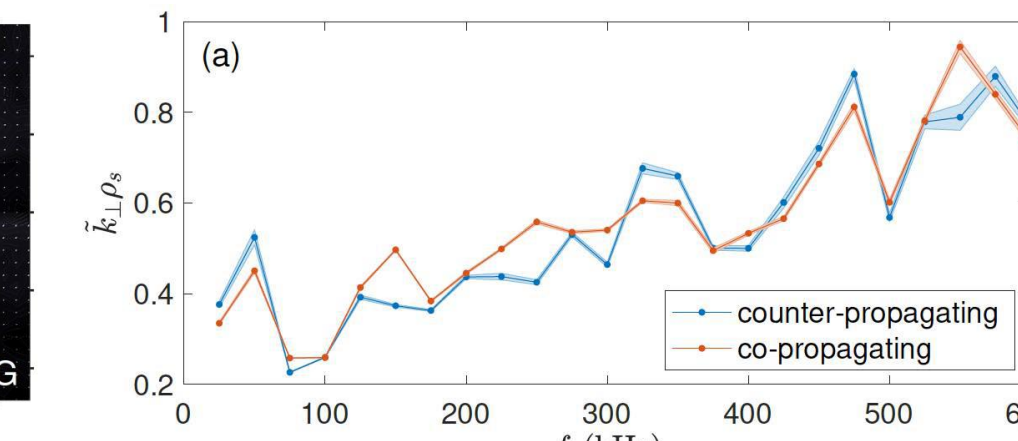
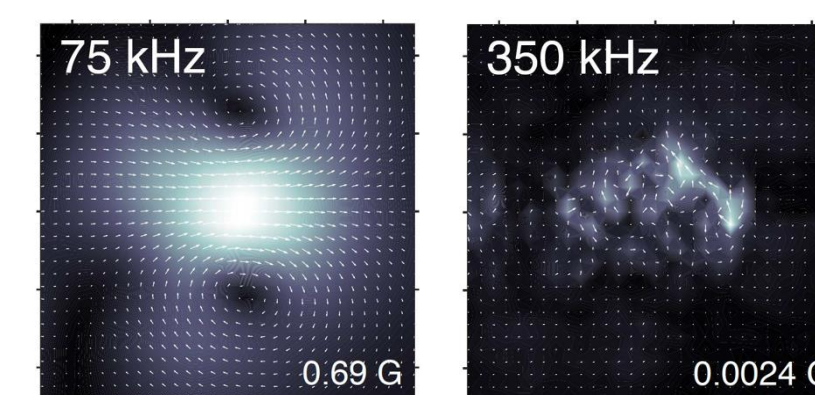
Lab Experiments



- Ran experiments at the UCLA Large Plasma Device (LAPD) to probe physics relevant to space plasma turbulence
- Showed that counter-propagating Alfvén waves generate spectrum of nonlinear (nonresonant) modes
- Also found co-propagating Alfvén waves produce similar spectrum, consistent with new finite $k_\perp d_i$ nonlinearity that may be relevant at high imbalance, e.g. near the Sun
- Showed generation of smaller perpendicular scales as interactions moved energy to higher frequencies, important step towards getting fully-developed turbulence in the lab



LAPD data: Spectrum of non-linear modes generated from interacting Alfvén waves in LAPD lab experiment and measurements showing generation of smaller scales as energy is transferred to higher frequency (Chen et al. 2026)



Open questions & approaches

Key open questions include:

- Determining the fundamental universal principles of the turbulent cascade
- Explaining all observed features in different parameter regimes
- How & why it varies throughout all the different heliophysics plasmas
- How it is driven and dissipated (i.e. processes that thermalise the energy)
- How it contributes to shaping large-scale dynamics / structure in the heliosphere

Future directions to help answer these questions:

- Observations: high-resolution low-noise fields, particles, and imaging data to probe small scale cascade, energy conversion & plasma heating; multi-spacecraft to understand full 3D structure; different locations to study new plasma regimes
- Techniques: new analysis methods driven by theory/simulations/observations
- Models: complete consistent fundamental turbulence models, and coupling between these and global system dynamics models
- Experiments: new lab experiments to probe regimes relevant to heliophysics plasmas, and create fully-developed turbulence
- Programmes: research programmes that link these aspects, turbulence focussed space mission(s), turbulence objectives in other missions, connections to astrophysics where plasma turbulence plays a role in important questions

Cross-disciplinary aspects

- Turbulence occurs throughout all heliophysics plasmas
- Brings together many approaches: highest resolution observations, largest scale simulations, novel nonlinear plasma theory, state of the art lab experiments
- Brings together communities: plasma physics, geophysics, fluid dynamics, space weather, lab plasma physics, astrophysics, space technology
- In particular, questions about how turbulence affects the systems in which it is in, e.g., our ability to predict space weather, are cross-cutting

